

Crop yield prediction using multi-attribute weighted tree-based support vector machine

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ARTICLE INFO

Keywords:

Agriculture
Crop productivity
Machine learning (ML)
Z-Score normalization
Principal component analysis (PCA)
Genetic algorithm (GA)
Multi-attribute weighted tree-based support vector machine (MAWT-SVM)

ABSTRACT

Agriculture is a prevalent and low-paying occupation in India. Agriculture is the key factor in the development of civilization. India is a predominantly agrarian nation with a crop-based economy. We can argue that agriculture can sustain the economy of our nation as a consequence. When organizing an agricultural enterprise, each crop must be properly picked. Numerous variables, such as market price, production volume, and governmental restrictions, will affect the crop selection. Numerous changes need to be made in the agriculture sector if economic growth in India is to be improved. The need for an effective technique to simplify crop cultivation and aid farm owners in crop management and production arises from the environment's unpredictable climate variables. This might enable potential farmers to practice sustainable agriculture. By modifying the economic position by producing the optimal crop, machine learning (ML) may lead to a revolution in the field of agriculture. To accurately predict crop yield, this research proposes the multi-attribute weighted tree-based support vector machine (MAWT-SVM) approach. Python software was used to conduct this study. Due to the existence of noisy data, the dataset for this study is first obtained, and the obtained raw data is normalized using the z-score normalization approach. Then, to extract the important features, principal component analysis (PCA) being employed. Additionally, the genetic algorithm (GA) employed to improve the suggested MAWT-SVM's performance by selecting the best features. Contrasted with other methods, the solution show that the suggested method achieves the best performance in crop yield prediction.

1. Introduction

Agriculture and allied businesses are a major source of income across the world, particularly in rural regions. The world's population is expanding, and unforeseen climate changes are occurring continuously to captivate farmers to find methods for enhancing agricultural productivity. This continuous endeavor to feed the growing masses is leading to the fast depletion of natural resources such as soil nutrients and groundwater levels. Scientists all over the world are now working on practices that can substantially increase crop production without much degradation of weather patterns and farming techniques, which protect our natural resources PS [1]. The need to protect the world's food supply has arisen as an outcome of climate change and the rising global population. With changing weather and global economic pressure, farmers must make tough choices on how to be productive and sustainable. Farmers may make the best choice for their cropping scenarios with the help of precise and timely information about weather,

soil, fertilizer usage, and pesticide use. If the circumstances are favorable, this might assist them to increase agricultural production or reduce loss brought on by unfavorable crop yield conditions Gandhi et al. [2]. A hundred different crops have been planted all over India. To facilitate comprehension & mainly characterized, these crops have been grouped. Accurate projections of crop production and timely may support the government in making tactical choices like arranging import and export, creating a strategic plan like crop pricing, and anticipating the prospect of famine and the prior estimation of crop yield can motivate the farmer in adopting sustainable agriculture practices on the farm, like rational use of water, promotion of soil health, and lowering of pollution levels, without worrying about the yield or returns from the field Van et al. [3]. Food security is crucial for the whole globe. Currently, there must be 7.6 billion humans on the planet, and it is predicted that by 2050, there will be 9.8 billion. The system for monitoring food security will face difficulties as a result of a 70 %–100 % increase in the need for food production. Currently, large-scale crop production assessments and

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<https://doi.org/10.1016/j.measen.2023.101002>

Received 18 February 2023; Received in revised form 12 May 2023; Accepted 19 December 2023

Available online 26 December 2023

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agricultural management both employ satellite remote sensing technologies Dang et al. [4]. Crop yield forecasting is a critical yet challenging issue that must be solved for enhancing agricultural productivity and the wise use of environmental assets. The various crop plant features, environmental factors, & management choices that affect crop output make it accurate and explicable. These techniques each target a different element of predicting agricultural yields Paudel et al. [5]. In this article, we propose the MAWT-SVM for crop yield prediction.

The remaining section of this article is structured as follows. We offer a thorough analysis of literature survey in Section 2. We outline the formulation of the proposed work in Section 3, and the results are shown and discussed in Section 4. The conclusions & recommendations for further research are offered in Section 5.

2. Literature survey

Nishant et al. [6] predicted the production of practically all crops that are cultivated in India. When those techniques were used together, the outcome was less innovative. Random Forests (RF), a ML technique, was examined by Jeong et al. [7] for its potential to forecast agricultural production responses to climate and biophysical factors at global and regional scales. As a result of its maximum quality & precision, simplicity of usage, & value while analyzing data, RF is a successful & flexible ML approach for agricultural production projections at local & global levels. Chlingaryan et al. [8] demonstrated how various ML approaches can handle these various but closely related jobs. The research comes to get around these restrictions; a variety of techniques may be used, including automated cross-validation, outlier identification, transfer learning, expert knowledge inclusion into the variance value, and model selection. Bondre and Mahagaonkar [9] examined the crop in several ways and suggested fertilizers appropriate for each crop. The conclusion is that RF has a better accuracy of 86.35 % when compared to Support Vector Machine for soil classification. Kumar et al. (2019) suggested addressing the crop selection issue, optimizing the net higher crop yield throughout the season, & ultimately achieving maximum economic development of the nation. Results demonstrate that, with very few exceptions, state-level agricultural production data and sub-division season are identical to the overall Indian outcome. According to Pant et al. [10], four popular yields that are mostly grown across India will be produced. According to the aforementioned findings, the decision tree linear regression model successfully matches the measurements to the maximum score. The models suggested by Shook et al. [11], various ML models that competed, including regression, Support Vector Generalized linear models with Radial Basis Function kernel, as well as the information model for crop yield prediction. The findings, which are based on more than a decade's worth of data, also show that prior-season weather factors are very few helpful in predicting seed production and that a proxy indicator is needed to support the genetic adaptability of soybean genotypes during earlier development and growth phases. The two crop support vector machines and the crop gradient boosting machine were suggested by Aworka et al. [12]. Results reveal that the 3 suggested ML models accurately predicted the crop information with an R^2 value of high precision. Using high-resolution information and mathematical method, Burdett and Wellen [13] analyzed the methods for quantifying the link between soil and topography parameters for forecasting crop output. These findings imply that if fields exhibit comparable connections between soil metrics and yield, ML approaches may be utilized to forecast high-yield regions of fields requiring existing yield maps. Nevavuor et al. [14] evaluated ML methods for remote sensing-based yield prediction require the necessity of yield mapping system, till now that are not usual between farmers. The findings demonstrate that the greatest data window size examined produced the lowest scoring errors. In the upcoming, the generated model will be refined for accuracy by training on a bigger collection of characteristics together with time series picture data. A supervised modern agricultural system built on the deep algorithm for

reinforcement learning was suggested by Elavarasan et al. [15]. The results of the productivity & precision tests demonstrate the effectiveness of the suggested Neural Recurring Q-Network for agricultural production prediction in a wide range of applications. The exciting possibility of concentrating on further computational efficiency enhancement in the training process. Batool et al. [16] examined the Food & Agriculture Organization of the United Nations Aqua Crop simulation model & techniques to analyze and compare the methodologies for predicting tea output. By keeping more thorough records of crop and field experimentation, the outcomes may be further enhanced. To assess how the fundamental algorithms perform in various performance metrics, Oikonomidis et al. [17] created deep learning-based models. The XGBoost model, which executed faster than the other DL-based models, was responsible for the outcome. A new chart recurrent neural network for agricultural yield prediction was presented by Fan et al. [18], to increase the model's prediction ability by including both spatial and temporal information. Researchers' evaluations of the effects of numerous variables on agricultural yields and temperature were examined by Bali et al. [19] and precipitation has been proven to be the greatest impact on the agricultural yields of various crops. As a result of the proposed neural network model's capacity to recognize & take use of the superfluous information included in soil and weather-related data.

3. Proposed methodology

In this part, we explain the suggested method of the multi-attribute weighted tree-based SVM for crop yield prediction. Fig. 1 illustrates the proposed methodology. The dataset is first gathered and prepared using the z-score normalization approach. PCA is used to extract the features, Feature Selection using a Genetic algorithm (GA), and a suggested multi-attribute weighted tree-based support vector machine is used for crop yield prediction.

3.1. Datasets

From 1999 to 2007, statistics on agricultural productivity and

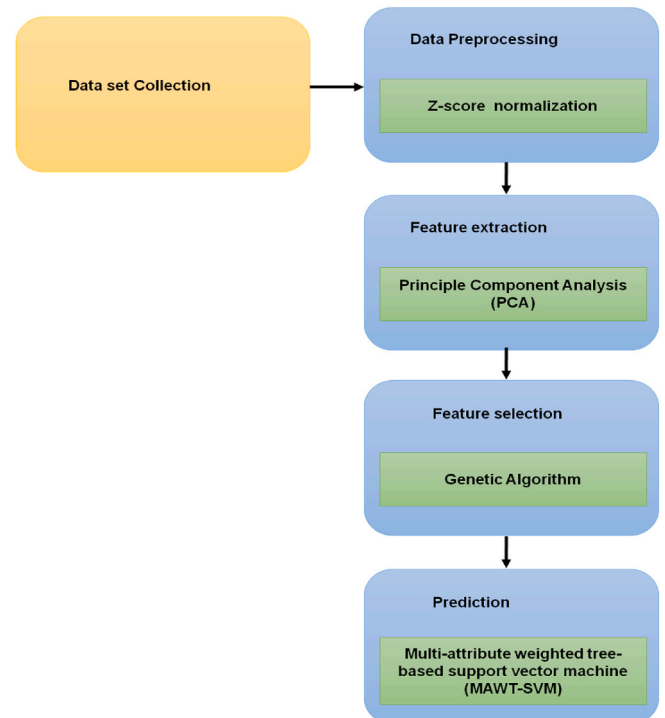


Fig. 1. Proposed methodology.

meteorological information were gathered from two different sources. The first set of information included characteristics of the sown areas, crop kinds, amount of irrigated water, beginning and finishing planting dates, and yield of crops. These data came from the Spriter-GIS system. Climate-related variables such as precipitation, solar output, and temperature are included in the second data source. The National Meteorological Service sites close to the module were used to gather weather information. Based on past CYP investigations and the availability of data, eight possible features (Table 1) were chosen. Since this study employs a full approach to identify the ideal characteristic subset for each logistic regression, these qualities are referenced as prospective attributes. The execution of the algorithm determines the final selection of characteristics. The last three phases of crop growth, which have the greatest influence on the crop's development, were used to determine the average of the weather's three main components: sun radiation, temperatures, and humidity.

Table 2 describes the crop datasets. Each of these datasets has an ID to make future referencing easier. The number of records and durations utilized for the phases of training and testing are shown in Table 2.

3.2. Preprocessing using z-score normalization

The values must be normalized for the data processing procedure. Only rescaling is required in certain methods of normalization to achieve values corresponding to a different variable. They must make a few simple modifications to the mistakes after we know the crop population characteristics. The population values may be distributed routinely rather than randomly once the mistakes have been corrected. Getting the z-score is the first step in the normalization procedure. The z-score is represented by eq. (1).

$$y = [(w - \mu) / \sigma] \quad (1)$$

Where μ denotes Median of the crop population & σ denotes variance of the crop population.

The standard score will be determined using the sample mean and sample standard deviation. If the means and standard deviation of the crop population are unknown, as shown in Eq. (2).

$$y = \frac{w - \hat{w}}{T_w} \quad (2)$$

Where w denotes the mean of sample and T_w denotes standard deviation of the sample.

The initial step in data research is pre-processing. The information will be converted into an understandable numerical representation. The elimination of priceless phrases is a key objective of this approach. The pre-processing stage would be simplified by uniformity. Mathematical encoding is the initial step in the integer value assignment process.

$$I = [(l - \mu) / \sigma] \quad (3)$$

Where μ represents mean of data amount and σ represent standard deviation of data. The normal integer value will be calculated by using the sample mean and average deviation when the data mean & standard deviation are unknown.

Table 1
Possible features in crop datasets.

Attribute code name	Attribute description
SP	Rainfall (mm)
SGR	Solar radiation (M-Joules/m ²)
IWD	Irrigation water depth applied (mm)
RH	Relative humidity in leaves (%)
Mint	Minimal temperature (°C)
MaxT	Maximal temperature (°C)

Table 2

Crop dataset test and training sample distribution.

Cultivar	Dataset ID	Crop species	Testing samples	Training period
Peruano	CBP02	Common bean (Phaseolus vulgaris)	361 2007 9	1999–2007
Alpha	PA06	Potato (Solanum tuberosum)	1749 2007 116	1999–2007
Mayocoba		Common bean (Phaseolus vulgaris)	332 2007 27	1999–2007
Azufrado	CBA03	Common bean (Phaseolus vulgaris)	120 2007 21	1999–2007
Jalapeno	PJ01	Pepper (Capsicum)	116 2006 18	1999–2006
Pioneer 30G54	CP05	Corn (Zea mays)	179 2006 19	2000–2007
Atlantic	PA07	Potato (Solanum tuberosum)	1062 2007 92	1999–2007
Saladette	TS08	Tomato (Lycopersicon esculentum Mill.)	182 2006 15	1999–2006

$$I = \frac{l - \hat{l}}{T_w} \quad (4)$$

$\hat{l}$ denotes mean of the sample and T_w denotes SD of the sample. The sequence may be transformed into a matrix format for suitable variable assignment.

$$E = I^* (I^S)^{-1} \quad (5)$$

The matrix's variance is,

$$\text{Var}(\hat{u}_j) = \sigma^2 (1 - u_{jj}) \quad (6)$$

$$\text{Var}(\hat{u}_j) = \sigma^2 \left(1 - \frac{1}{j} - \left[(l_j - l^2) / \sum_{i=1}^j (l_i - l^2) \right] \right) \quad (7)$$

Next, the residual that may be determined by

$$\frac{u_j}{\sigma} \sqrt{1 - u_{jj}} \quad (8)$$

The function scaling approach may then be used to start all variable values at 0. Integer encoding is the term used to describe this.

$$Y' = \frac{(I - c_{\min} \text{Var}(c_j) l_j)}{(c_{\max} - c_{\min}) \sigma^2} \quad (9)$$

3.3. Feature extraction using principal component analysis

A selection model has been developed to choose the suitable crops to be grown in the specified experimental field. The 26 input variables that were chosen and categorized are divided into 6 main categories: land, water, season, input, profit, & yield. 26 input variables were obtained throughout the feature extraction process to construct the decision-making model that particular agricultural region. Using the feature selection approach, Principal Component Analysis selects six of the twenty-six input variables. The PCA system may then be used to visualize the characteristics. In this PCA algorithm map, the parameter value for each class may be shown. By improving interpretability, PCA is a method for decreasing dataset dimensionality. By using a form of transformation theory, PCA reduces data from many different variables into a smaller number of variables. When there is a correlation, it suggests that the material is repetitious and, if it is constant, the information may be condensed. Taking into account the F1 and F2 characteristics, these are equally distributed,

The O output class and a [1, 1] binary, are both provided below.

$$P = \begin{cases} 0 & \text{if } E1 + E2 < 0 \\ 1 & \text{if } E1 + E2 \geq 0 \end{cases} \quad (10)$$

The data points in this conundrum are shown in murky areas. The issue may be divided linearly, and picking the necessary characteristics of E1 + E2 is simple. On a group of data providers, the LDA is carried out using N-dimensional vectors, which show the direction of the feature space. This vector gives the output class that projects the issue into space the most up-to-date function and the most accurate information about the issue (E1, E2).

For every single class, the scatter matrix for the T_{ot} class and the T_{lq} class, which are specified as follows, will be used to create the matrix:

$$T_{ot} = \sum_{b=1}^{oT} T_b; T_b = \frac{1}{o_i} \sum_{o \in o_i} (o - k_b)(o - k_b)^S \quad (11)$$

$$T_{lq} = \sum_{b=1}^{LQ} (k_b - k)(k_b - k)^S \quad (12)$$

$$O = \frac{1}{m} \sum_{o \in O} (o - 1)(o - 1)^S \quad (13)$$

It is possible to sort the pointed features by getting the covariance matrix.

3.4. Feature selection using genetic algorithm

Genetic algorithms are based on one natural phenomenon, evolution. Fig. 2 depicts the genetic algorithms process. Three operators are most common: Selection Two members of the current population are chosen at random throughout each succeeding generation to produce a new person. The selection procedure often takes into account a person's level of fitness. The roulette game is a common form of selection, and fitter people are favored. There is a compelling reason why the kid solution that incorporates the elements of a successful parent solution is more likely to be successful. When the community is huge, this procedure might take a while. Additionally, some techniques choose the parent solutions at random. We do not have to determine each solution's fitness while using these approaches. Crossover The crossover operator is used after the choice of the parent solutions. Another name for crossover is

recombination. To create a new unique solution, it is employed to combine the favorable genes of the parent solution. Replace the old solutions with the new ones if the created solution is superior to the parent solutions.

The crossover operator's essential feature, particularly for the huge optimization issue, is the ability to identify the favorable genetics of the parent solutions. Shrinkage of the parent solutions' exchange scope is one remedy for this. Mutation The mutation is a biological operator used in genetic algorithms to preserve genetic diversity. There is no new gene added throughout the selection process and crossover. In contrast, some genes may be lost in the process. This has the effect of making every member of the population similar, which causes the algorithms to converge quickly. The mutation is employed to randomly change one or more genes to prevent this scenario. Since the mutation operator uses a random search methodology, the preceding result could significantly alter. As a result, the mutation operator often has a low threshold probability of execution.

3.5. Multi-attribute weighted tree-based support vector machine

The occurrence of unclassifiable areas may be eliminated using Multi-attribute weighted tree-based SVM, which also has a stronger generalization capability than the traditional approach. Two instances of feature space division and the accompanying decision tree are shown in Figs. 3 and 4. Different tree structures correspond to different ways of dividing up the feature space, and the relationship between the multi-attribute weighted tree-based structure & the classifier's categorization effectiveness is strong. Figs. 3 and 4 shows example 1 and example 2.

Training is the name of the procedure used to create the decision tree. Except for the leaf node, every node is connected a subspace dividing one class from another. The top node of decision tree's subspace must first be identified. The hyper plane separating the divided categories at the node connecting to the top node should be found if there are many classes in the divided categories. Only one class should remain in the isolated area when this operation is completed. When a leaf nodes is reached in the classification process, the input x may then be placed into the category that the node denotes. The method of classification begins at the top of multi-attribute weighted tree & requires calculation of the value of the surface x for the raw data. The next node is determined by the surface of the existing node's value. In this approach, the multi-attribute weighted tree-based structure directly affects how well the classifier does classification, thus it's critical to research how to choose the decision tree's structure based on training samples to reduce classification error. If the categorization performance is poor at the decision tree's top node, the classification performance as a whole degrades. As a result, the multi-attribute weighted tree-based higher node should divide more separable types.

4. Result and discussion

Machine learning frameworks were used extensively in this study's experimentation, which was conducted on a personal DELL laptop with an Intel Core i3-4300U CPU running at 1.90 GHz and 8 GB of RAM. In this part, the significance of the project's outcomes is addressed. 30% of data will be used for testing, & 70% for training. In beginning, we are selecting samples for the model's training and select characteristics like yield, temperature, and location, among others. Following the dataset's successful testing and training, we continued our search for the model's precision. The more closely the model's accuracy resembles the actual yield number, the better we were able to evaluate the crop's yield when compared to the original data. By illustrating the yield variable with various properties, such as Accuracy, Precision, Recall, Dice coefficient, and Jaccard coefficient.

• Accuracy

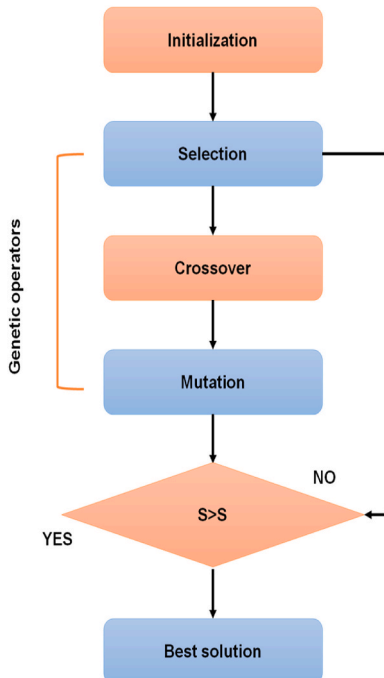


Fig. 2. Workflow of genetic algorithm.

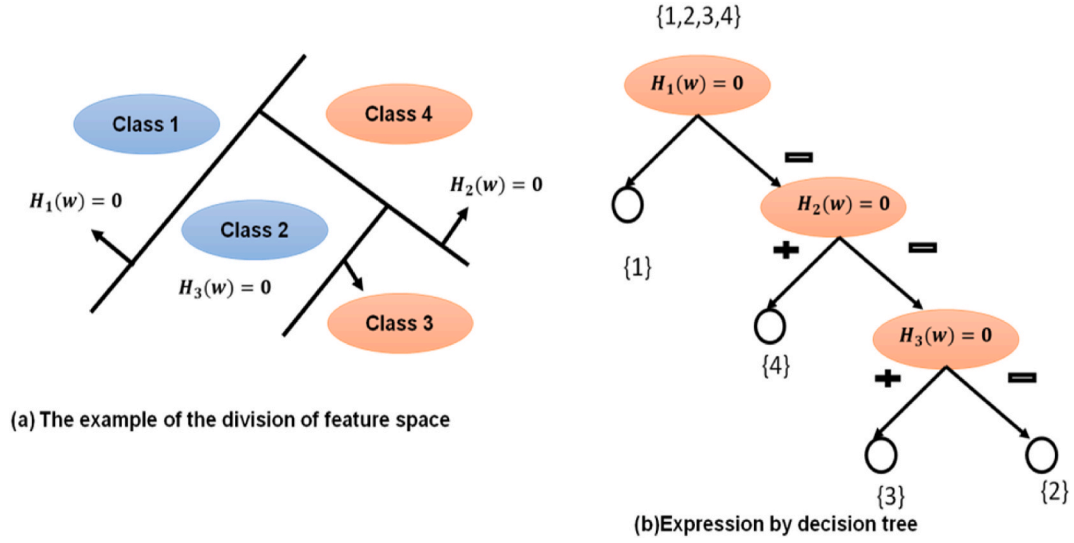


Fig. 3. The segmentation of feature space & related decision tree, example 1.

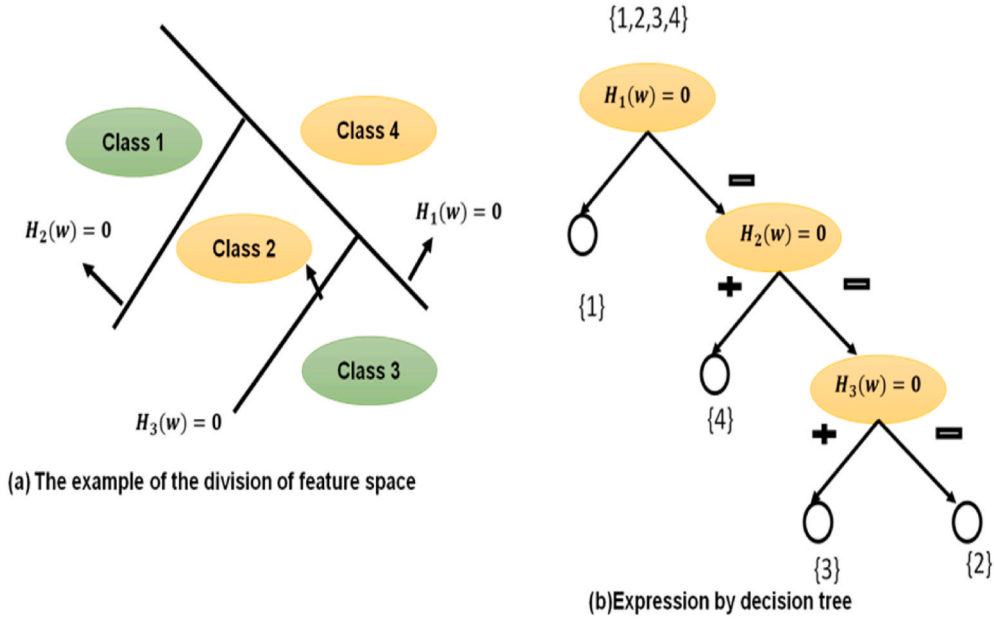


Fig. 4. The segmentation of feature space & related decision tree, example 2.

This plot of structured error serves as a propensity indicator; low accuracy results in a discrepancy between the result and the real value. In other cases, the same procedure is used to examine the value, and the actual results of the model that was applied are analyzed. The proportion of real results indicates how well-rounded the data is overall. The accuracy is calculated using equation (14).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

Fig. 5 displays comparable values for the accuracy metrics and makes it evident that the suggested technique can generate performance results that are better than those generated by the current research approaches. The suggested method's accuracy of 98.5 % outperforms the results of the current ones, which include CNN having a 93 % accuracy rate, DNN at 83 %, and MLR-ANN at 72 %.

• Precision

A representation of random error that measures algebraic variability is called precision.

Precision is calculated using equation (15).

$$precision = \frac{TP}{TP + FP} \quad (15)$$

Fig. 6 displays comparable values for the precision measures. With a 97 % precision, the suggested method outperforms the currently used ones, which include CNN in precision (70 %), DNN (85 %), and MLR-ANN (89 %) with a precision of 92.34 %. The proposed method achieved more data categorization precision than other methods currently in use.

• Recall

In certain sectors, recall accurately specifies the genuine optimism rate, warning, or chance of identification. It also estimates the percentage of positive facts. The recall is calculated using equation (16).

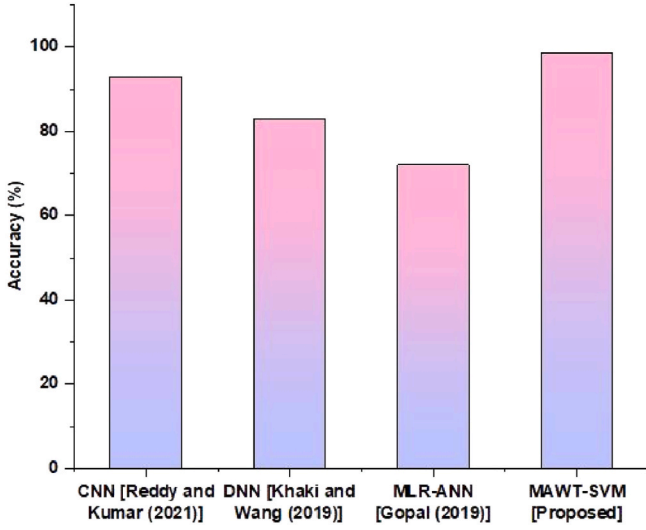


Fig. 5. Results of accuracy.

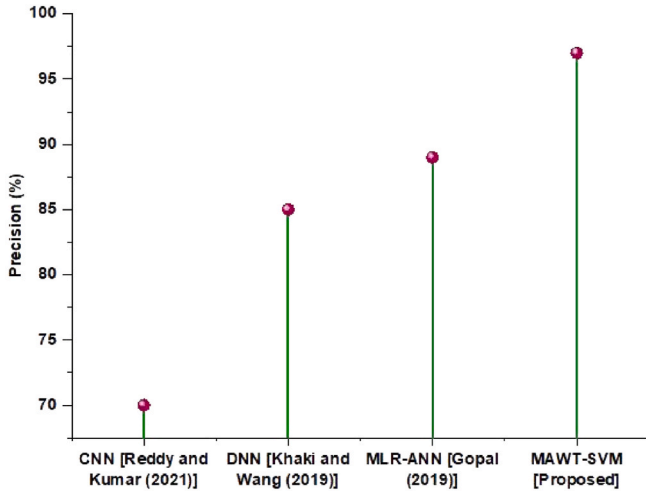


Fig. 6. Results of precision.

$$\text{Recall} = \frac{\text{FN}}{\text{FN} + \text{TP}} \quad (16)$$

Comparative data for the recall metrics are shown in Fig. 7. With a 95 % recall, the suggested method outperforms the currently used ones. Recall rates for CNN are 92 %, DNN 83 %, MLR-ANN 70 %.

• Dice coefficient

We suppose that the data characteristics discovered and the attributes of the basic truth data, respectively, are p and v . The Dice coefficient is calculated using equation (17).

$$T(p, v) = (2p \cap v / e + p) = 2DB / 2DB + LM + LB \quad (17)$$

Comparative data for the dice coefficient are shown in Fig. 8. With a 98.5 % dice coefficient the suggested method outperforms the currently used ones, the dice coefficient for CNN is 76 %, DNN 81 %, MLR-ANN 92 %.

• Jaccard coefficient

The calculation for the two groups' similarity & the Jaccard coefficient is calculated using equation (18).

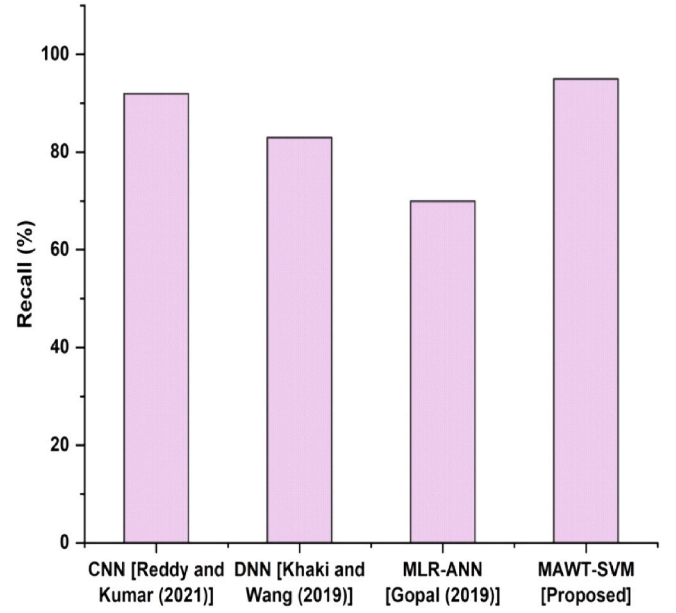


Fig. 7. Results of recall.

$$I(p, v) = (p \cap v / p \cup v) = (p \cap v) / p + v - (p \cap v) = (DB / DB + LM + LB) \quad (18)$$

Comparative data for the Jaccard coefficient are shown in Fig. 9. With a 98.6 %, the suggested method outperforms the currently used ones; Jaccard coefficient for CNN is 81 %, DNN 86 %, MLR-ANN 91 %.

Regarding the solution, Figs. 8 and 9 demonstrates that the suggested approach accurately displays big Jaccard values & the dice coefficient.

5. Conclusion

The agricultural sector is the life of countries like India. However, precision agriculture must place a high priority on the use of technology in agriculture. In this research, we provide machine learning approach known as MAWT-SVM for crop yield prediction. The dataset is first gathered and prepared using the z-score normalization approach. We use Principle Component Analysis (PCA) for feature extraction and Genetic Algorithm for feature selection. Performance indicators for multi-attribute MAWT-SVM include accuracy (98.5%), precision (97%), recall (95%), Dice coefficient (98.5%), Jaccard co-efficient (98.6%). As a result, when compared to previous approaches, our suggested system, which uses a MAWT-SVM, performs better. Sustainable farming is typically aided by yield preservation strategies that maximize resource use. It's possible that restricting the use of agrochemicals and other strategies for protecting agricultural yields might have unfavorable effects on the environment. In the future, we'll create a platform that enables users to investigate crop production prediction across a range of crops utilizing dependent and independent factors. Research and development in the future will concentrate on perfecting and enhancing these techniques.

Code availability

Not applicable.

Ethics approval

Not applicable.

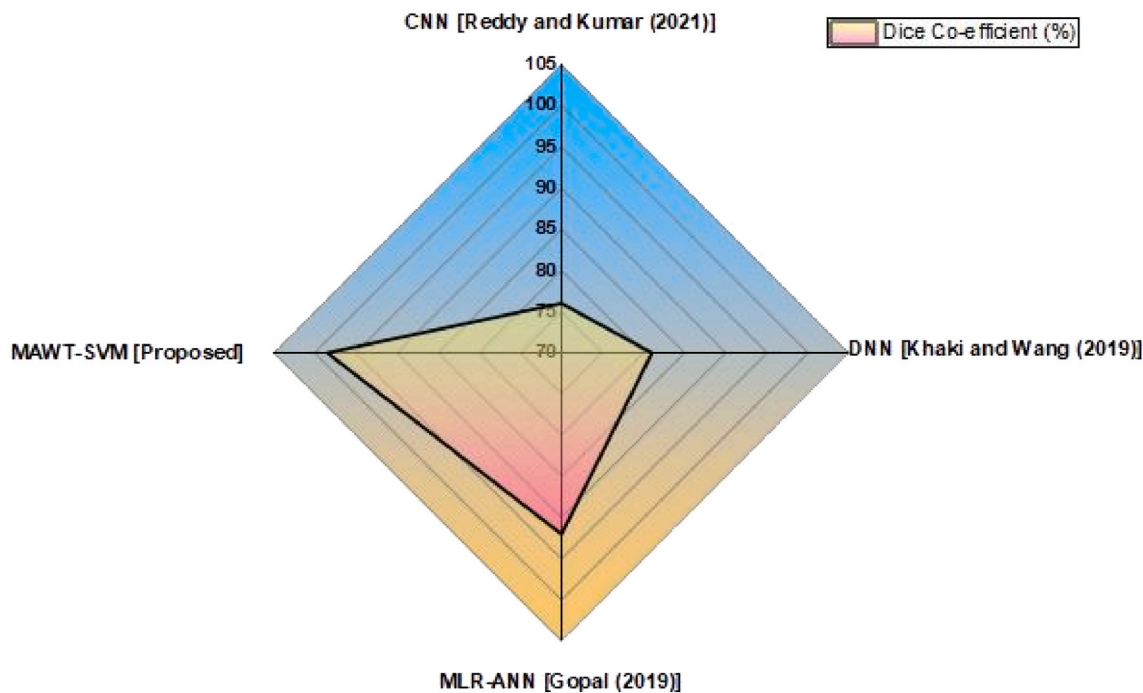


Fig. 8. Results of Dice coefficient.

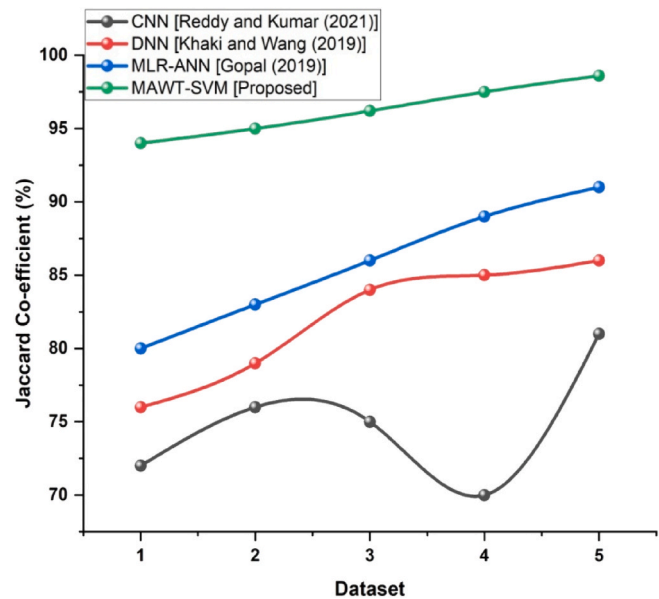


Fig. 9. Results of the Jaccard coefficient.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

There is no funding

References

- [1] M.G. PS, Performance evaluation of best feature subsets for crop yield prediction using machine learning algorithms, *Appl. Artif. Intell.* 33 (7) (2019) 621–642.
- [2] N. Gandhi, L.J. Armstrong, O. Petkar, A.K. Tripathy, July. Crop yield prediction in India using support vector machines, in: 2016 13th International Joint Conference on Computer Science and Software Engineering (JCSSE), IEEE, 2016, pp. 1–5.
- [3] T. Van Klompenburg, A. Kassahun, C. Catal, Crop yield prediction using machine learning: a systematic literature review, *Comput. Electron. Agric.* 177 (2020), 105709.
- [4] C. Dang, Y. Liu, H. Yue, J. Qian, R. Zhu, Autumn crop yield prediction using data-driven approaches: support vector machines, random forest, and deep neural network methods, *Can. J. Rem. Sens.* 47 (2) (2021) 162–181.
- [5] D. Paudel, H. Boogaard, A. de Wit, S. Janssen, S. Osinga, C. Pylianidis, I. N. Athanasiadis, Machine learning for large-scale crop yield forecasting, *Agric. Syst.* 187 (2021), 103016.
- [6] P.S. Nishant, P.S. Venkat, B.L. Avinash, B. Jabber, June. Crop yield prediction based on Indian agriculture using machine learning, in: 2020 International Conference for Emerging Technology (INCET), IEEE, 2020, pp. 1–4.
- [7] J.H. Jeong, J.P. Resop, N.D. Mueller, D.H. Fleisher, K. Yun, E.E. Butler, D.J. Timlin, K.M. Shim, J.S. Gerber, V.R. Reddy, S.H. Kim, Random forests for global and regional crop yield predictions, *PLoS One* 11 (6) (2016), e0156571.
- [8] A. Chlingaryan, S. Sukkarieh, B. Whelan, Machine learning approaches for crop yield prediction and nitrogen status estimation in precision agriculture: a review, *Comput. Electron. Agric.* 151 (2018) 61–69.
- [9] D.A. Bondre, S. Mahagaonkar, Prediction of crop yield and fertilizer recommendation using machine learning algorithms, *International Journal of Engineering Applied Sciences and Technology* 4 (5) (2019) 371–376.
- [10] J. Pant, R.P. Pant, M.K. Singh, D.P. Singh, H. Pant, Analysis of agricultural crop yield prediction using statistical techniques of machine learning, *Mater. Today: Proc.* 46 (2021) 10922–10926.
- [11] J. Shook, T. Gangopadhyay, L. Wu, B. Ganapathysubramanian, S. Sarkar, A. K. Singh, Crop yield prediction integrating genotype and weather variables using deep learning, *PLoS One* 16 (6) (2021), e0252402.
- [12] R. Aworka, L.S. Cedric, W.Y.H. Adoni, J.T. Zoueu, F.K. Mutombo, C.L.M. Kimpolo, T. Nahhal, M. Krichen, Agricultural decision system based on advanced machine learning models for yield prediction: a case of east African countries, *Smart Agricultural Technology* 2 (2022), 100048.

- [13] H. Burdett, C. Wellen, Statistical and machine learning methods for crop yield prediction in the context of precision agriculture, *Precis. Agric.* (2022) 1–22.
- [14] P. Nevavuori, N. Narra, T. Lipping, Crop yield prediction with deep convolutional neural networks, *Comput. Electron. Agric.* 163 (2019), 104859.
- [15] D. Elavarasan, P.D. Vincent, Crop yield prediction using deep reinforcement learning model for sustainable agrarian applications, *IEEE Access* 8 (2020) 86886–86901.
- [16] D. Batool, M. Shahbaz, H. Shahzad Asif, K. Shaukat, T.M. Alam, I.A. Hameed, Z. Ramzan, A. Waheed, H. Aljuaid, S. Luo, A Hybrid approach to tea crop yield prediction using simulation models and machine learning, *Plants* 11 (15) (2022) 1925.
- [17] A. Oikonomidis, C. Catal, A. Kassahun, Hybrid deep learning-based models for crop yield prediction, *Appl. Artif. Intell.* 36 (1) (2022), 2031822.
- [18] J. Fan, J. Bai, Z. Li, A. Ortiz-Bobea, C.P. Gomes, June. A GNN-RNN approach for harnessing geospatial and temporal information: application to crop yield prediction, *Proc. AAAI Conf. Artif. Intell.* 36 (No. 11) (2022) 11873–11881.
- [19] N. Bali, A. Singla, Emerging trends in machine learning to predict crop yield and study its influential factors: a survey, *Arch. Comput. Methods Eng.* (2022) 1–18.